packages/graph-view/src/utils/selectionCluster.ts

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Overview

A new file `packages/graph-view/src/utils/selectionCluster.ts` (lines R1-87) adds utilities for working with graph nodes and file paths. It imports `GraphEdge` and `KnowledgeGraph` from `@novadiff/graph-core/types` (R1) and defines a set of link-relevant edge types (R3).

Key changes

- **`filePathFromNodeId`** (R5-14) parses a node ID to extract a file path for `file:`, `function:`, or `class:` prefixes.
- **`resolveGraphNodeFilePath`** (R17-29) returns a node's `filePath` from the graph or falls back to `filePathFromNodeId`.
- **`graphNodeIdsInFile`** (R32-49) collects all symbol and file node IDs belonging to a given file path, deduplicating with a `Set`.
- **`graphSelectionCluster`** (R56-87) builds a cluster that includes the target node, all nodes in the same file, and one-hop neighbors over `imports`, `calls`, or `contains` edges.
- The `LINK_EDGE_TYPES` set (R3) filters edges used in the cluster calculation.

Impact

- **Correctness**: deterministic file-path resolution and cluster calculation.
- **Maintainability**: centralizes node-path logic, reducing duplication.
- **Performance**: linear scans over `graph.nodes` and `edges`; caching could be added if profiling shows bottlenecks.
- **Compatibility**: no existing API changes; the new file only adds exports.

Risks & follow-ups

- **Parsing assumptions**: verify that all node ID formats in the repo match the patterns handled by `filePathFromNodeId`.
- **Edge filtering**: ensure `LINK_EDGE_TYPES` includes all edge types that should contribute to a selection cluster.
- **Duplicate handling**: `graphNodeIdsInFile` uses a `Set`; confirm this matches downstream expectations.
- **Test coverage**: add unit tests for each helper, covering edge cases such as missing `filePath` or empty graphs.